

CCNA 200-301 - OSPF Complete Notes

Overview

- Open Shortest Path First (OSPF)
- Link-State & Standard Protocol
- Classless (Supports VLSM/CIDR)
- Administrative Distance = 110
- Metric = Cost = Reference Bandwidth / Interface Bandwidth (Default Reference = 10^8)
- Multicast:
 - 224.0.0.5 (All OSPF Routers)
 - 224.0.0.6 (DR/BDR)
- Unlimited Hop Count
- Fast Convergence

OSPF Operation

- Hello Packets -> Discover Neighbors
- Exchange LSA
- Flood LSAs
- Build LSDB
- Run SPF (Dijkstra)
- Build Routing Table

Areas

Area 0 = Backbone Area

ABR connects Areas

ASBR connects OSPF with external networks.

DR / BDR Election

Priority (0-255, default 1)

Then Router ID.

Router ID order:

- 1- Manual router-id
 - 2- Highest Loopback IP
 - 3- Highest Physical Interface IP
- Priority 0 = Never DR/BDR

Configuration Commands

- router ospf 1
- router-id 1.1.1.1

- network 10.0.0.0 0.255.255.255 area 0
- interface loopback0
- ip address 50.0.0.1 255.255.255.255
- interface g0/0
- ip ospf priority 200
- no router ospf 1
- clear ip ospf process

Show Commands

- show ip ospf
- show ip ospf neighbor
- show ip ospf interface
- show ip ospf interface brief
- show ip ospf database
- show ip protocols
- show ip route
- show ip route ospf
- show running-config
- show running-config | section ospf

Important Notes

- Wildcard Mask = Inverse Subnet Mask.
- Process ID is locally significant.
- OSPF exchanges LSAs instead of full routing tables.
- Cost depends on bandwidth.
- Broadcast networks elect DR and BDR.
- Point-to-Point networks do not elect DR/BDR.
- Loopback keeps Router ID stable.